

Data Visualization using Matplotlib and Seaborn

Aim

To create and analyze different types of visualizations using Matplotlib and Seaborn libraries.

Scenario

Analyzing ShopKart Sales Performance

SCENARIO:

ShopKart Analytics Pvt. Ltd. collects sales information such as **month, product, sales amount, quantity, and number of customers**. The management wants to understand sales trends, compare products, analyze customer purchases, and identify patterns in the sales data.

As a **Data Analyst**, your task is to visualize and analyze the ShopKart sales data using Matplotlib and Seaborn.

Definition

Matplotlib and Seaborn are Python libraries used for data visualization. Matplotlib provides basic and customizable charts, while Seaborn provides simple and attractive statistical visualizations for analyzing patterns and relationships in data.

Functions Covered

Matplotlib

- `plot()`
- `bar()`
- `scatter()`

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Matplotlib and Seaborn are Python libraries used for data visualization. Matplotlib provides basic and customizable charts, while Seaborn provides simple and attractive statistical visualizations for analyzing patterns and relationships in data.

Functions Covered

Matplotlib

- plot()
- bar()
- scatter()
- hist()
- pie()
- xlabel()
- ylabel()
- title()

Seaborn

- lineplot()
- barplot()
- countplot()
- boxplot()

Student Task

Create a ShopKart sales dataset containing:

- Month

Student Task

Create a ShopKart sales dataset containing:

- Month
- Product
- Sales
- Quantity
- Customers

Perform the following:

Using Matplotlib:

1. Create a **line chart** to show monthly sales.
2. Create a **bar chart** to compare product-wise sales.
3. Create a **scatter plot** to show the relationship between customers and sales.
4. Create a **histogram** to show sales distribution.
5. Create a **pie chart** to show product-wise sales contribution.

Using Seaborn:

6. Create a **line plot** to show the sales trend.
7. Create a **bar plot** to compare average sales by product.
8. Create a **count plot** to show the number of purchases for each product.
9. Create a **box plot** to analyze the distribution of sales by product.

